

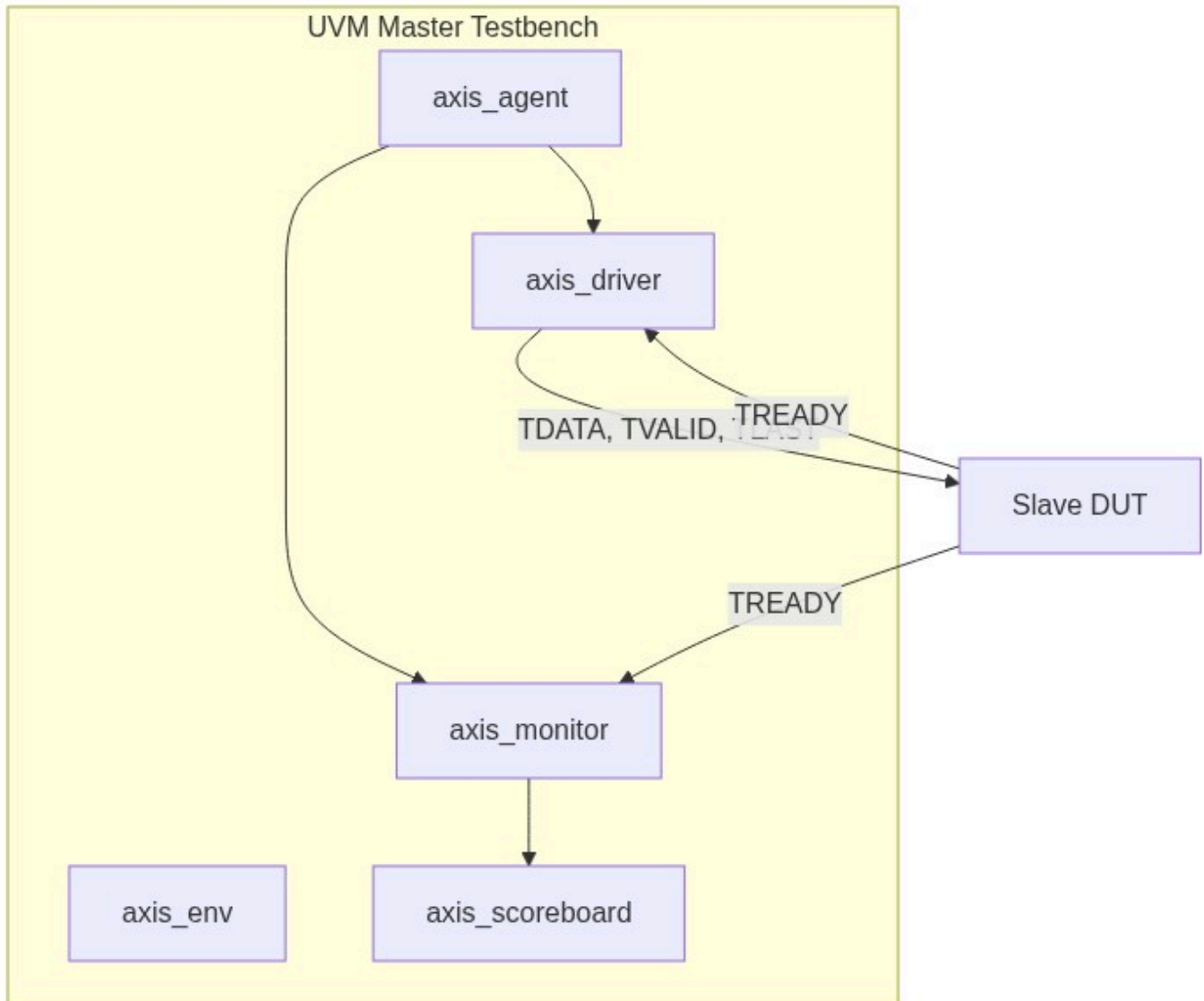
Verification Plan Report

Project: AXI-Stream Enterprise v3.2

Version: 3.2

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MS_01	Handshake (Stability)	Data Stability	Master VIP shall keep TDATA stable while observing TREADY=0 from Slave DUT.

Verification Plan Report

REQ_MS_02	Handshake (Latency)	TVALID Independence	Master VIP must not require TREADY to be asserted before it asserts TVALID.
REQ_MS_03	Reset (Timing)	Reset Suppression	Master VIP must drive TVALID to 0 during ARESETn.
REQ_MS_04	Reset (Recovery)	Reset Recovery	Master VIP must wait 1+ cycles after reset deassertion before driving TVALID.
REQ_MS_05	Packet (ID)	TID Integrity	Master VIP must hold TID constant throughout a packet until TLAST pulse.
REQ_MS_06	Packet (DEST)	TDEST Integrity	Master VIP must hold TDEST constant throughout a packet.
REQ_MS_07	Packet (Boundary)	TLAST Rule	Master VIP must assert TLAST on the final beat of every packet.
REQ_MS_08	Qualifiers (TKEEP)	Null Tracking	Master VIP TKEEP bits correctly identify data vs null bytes.
REQ_MS_09	Qualifiers (TSTRB)	Strobe Tracking	Master VIP TSTRB bits identify data vs position bytes.
REQ_MS_10	Power (AXI5)	Wakeup Protocol	Master VIP MUST hold TWAKEUP asserted if TVALID is waiting for TREADY.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HS_01	Baseline Handshake: Master VIP asserts TVALID. Observer sees Slave DUT TREADY=1. Handshake in 1 cycle.	REQ_MS_01, REQ_MS_02
TC_HS_02	Back-Pressure: Master VIP asserts TVALID. Observe TREADY=0 for 50 cycles. Verify TDATA remains frozen.	REQ_MS_01
TC_HS_03	Master Independence: Verify TVALID occurs even when TREADY is observed at constant 0.	REQ_MS_02
TC_HS_04	Back-to-Back: Mastery throughput with continuous TVALID/TREADY confluence.	REQ_MS_01
TC_HS_05	Master Stall: Master VIP deasserts TVALID mid-burst while Slave DUT observes TREADY=1.	REQ_MS_02
TC_RST_01	Async Reset: Assert reset mid-transfer. Verify Master VIP drives TVALID=0 immediately.	REQ_MS_03
TC_RST_02	Cold Recover: Drive TVALID on same cycle as reset release. Verify protocol violation check.	REQ_MS_04
TC_RST_03	Delayed Start: Wait 10 cycles after reset. Verify Master VIP resumes clean handshake.	REQ_MS_04

Verification Plan Report

TC_PKT_01	Minimal Packet: Single beat with TLAST=1.	REQ_MS_07
TC_PKT_02	Exhaustive Lengths: Packets of size 2, 8, 33, 64, 1024. Verify TLAST on final cycle.	REQ_MS_07
TC_PKT_03	TID Switch: Send Packet 1 (TID=0xA), then Packet 2 (TID=0xB) with zero gap.	REQ_MS_05
TC_PKT_04	TID Mid-Packet Violation: Attempt to toggle TID bits while TLAST=0. Trigger assertion.	REQ_MS_05
TC_PKT_05	TDEST Check: Verify TDEST routing for multi-destination slave configurations.	REQ_MS_06
TC_PKT_06	Zero-Byte Packet: TLAST=1 but all TKEEP bits=0. Verify Slave DUT acknowledge.	REQ_MS_08
TC_QUA_01	All-Ones TKEEP: Standard dense data transfer.	REQ_MS_08
TC_QUA_02	Sparse TKEEP: Drive packet where only even bytes are valid.	REQ_MS_08
TC_QUA_03	Data-Position Swap: Toggle TSTRB while TKEEP is 1 to verify position beats.	REQ_MS_09
TC_QUA_04	Parity Integrity: Inject AXI5 parity errors on TDATACHK bits.	REQ_MS_01
TC_PWR_01	Wakeup Assert: TWAKEUP asserted 1 cycle before TVALID. Observe Slave DUT response.	REQ_MS_10
TC_PWR_02	Wakeup Wait: Hold TWAKEUP and TVALID. Observe Slave DUT ready latency.	REQ_MS_10
TC_PWR_03	Illegal Wakeup Drop: Deassert TWAKEUP while TVALID=1 and TREADY=0.	REQ_MS_10
TC_PRF_01	Max Throughput: 10,000 beats with zero stall from both sides.	REQ_MS_01
TC_PRF_02	Intermittent back-pressure: Master VIP throttles if Slave DUT TREADY latency > 10 cycles.	REQ_MS_10
TC_STR_01	Random Handshake: Randomize TVALID gaps and observe Slave DUT random TREADY gaps.	REQ_MS_01, REQ_MS_10
TC_STR_02	Packet Size Jitter: Randomize length from 1 to 4096 beats.	REQ_MS_07
TC_STR_03	ID Pressure: Rapidly cycle all 256 TID values back-to-back.	REQ_MS_05
TC_STR_04	User Sideband Stress: Randomized bit-flipping on TUSER bits.	REQ_MS_01
TC_ADD_01	Unaligned Start: TKEEP[0]=0 at packet start. Verify Receiver handling.	REQ_MS_08
TC_ADD_02	Unaligned End: TKEEP[7]=0 at TLAST. Verify padding logic.	REQ_MS_08
TC_ADD_03	Multiple Terminations: Verify TLAST behavior in merged vs split streams.	REQ_MS_07
TC_ADD_04	Deadlock Timeout: Monitor observes TREADY stall exceeding 1,000,000 cycles.	REQ_MS_01
TC_ADD_05	Reset Recovery Race: Assert reset precisely on first TVALID beat.	REQ_MS_03
TC_ADD_06	TID Alias: Send different packets with identical TID but distinct TDEST.	REQ_MS_06

Verification Plan Report

TC_ADD_07	Illegal Strobe: Drive TSTRB without TKEEP. Observe VIP monitor checking.	REQ_MS_09
TC_ADD_08	Protocol Wrap: Run directed sequence of all corner cases in one simulation.	REQ_MS_10

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of Master TVALID and Observed Slave TREADY.
cg_packet_size	Distribution of beat counts per packet.
cg_reset_recovery	Delay from ARESETn rising edge to TVALID rising edge.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Output	TD_W	Data payload.
TVALID	Output	1	Master valid.
TREADY	Input	1	Slave ready (Observed).
TLAST	Output	1	Packet end.
TID	Output	TID_W	Stream ID.
TDEST	Output	TDEST_W	Dest ID.
TKEEP	Output	TD_W/8	Null bytes.
TSTRB	Output	TD_W/8	Strobes.

6. Verification Environment Structure

verif/axis_v3_2/

```
|-- agent/
|   |-- axis_master_driver.sv
|   |-- axis_master_monitor.sv
|   |-- axis_master_agent.sv
|   |-- axis_if.sv
|-- env/
|   |-- axis_env.sv
|   |-- axis_scoreboard.sv
|   |-- axis_pkg.sv
|-- tests/
|   |-- base_test.sv
|   |-- stress_test.sv
```

7. SystemVerilog Implementation

Verification Plan Report

Master Protocol Monitor (Role-Safe Observation)

```
if (vif.TVALID && vif.TREADY) begin
    // Monitor observes successful handshake from both sides
    axis_item tr = axis_item::type_id::create("tr");
    tr.data = vif.TDATA;
    tr.last = vif.TLAST;
    ap.write(tr);
end else if (vif.TVALID && !vif.TREADY) begin
    // Monitor observes protocol stall - verify stability
    `uvm_error("STABILITY_ERR", "TDATA changed during TREADY stall") if (vif.TDATA != last_tdata);
end
```